Ukrainian Catholic University

Faculty of Applied Sciences

Computer Science Program

Geospatial System for GPS-Independent Navigation Planning

Thesis Proposal

Author:
Maksym Mykhasyuta
Supervisor:
Vasyl Mylko

01 December 2024

Background and Motivation

Almost all modern aerial vehicles rely on GPS for navigation. Failures in GPS due to signal interference, jamming, or other technical issues can compromise safety. In such cases, alternative navigation methods that use visual geospatial features become critical. Current solutions often rely on specific object recognition, which requires significant computational resources and may falter in dynamic environments. It is much better to have some distinct unique features than only one or two unique objects for reference.

This research draws inspiration from urban morphology and street network analysis. Louf and Barthelemy's classification of street patterns[3] offers a macro-level understanding of urban layouts, while Boeing's entropy metrics [2] provide a framework for quantifying spatial order and complexity. These approaches help identify geospatially distinctive regions, such as settlements, infrastructure, and shaped parks with numerous unique objects, that could serve as potential reference points in GPS-denied scenarios.

By leveraging OpenStreetMap (OSM) data and geospatial analysis techniques[1], this research aims to determine the feasibility of identifying such regions and to analyze their distribution across selected areas. This study bridges the gap between theoretical urban studies and practical geospatial needs in aviation.

Problem Formulation

Small autonomous and man-operated aircraft lack a robust framework for pre-identifying geospatially distinctive regions that could serve as reference points in the absence of GPS. This research focuses on the feasibility of identifying such characterized by high visual distinctiveness regions and analyzing their distribution across a country or administrative area. These insights could be used in future projects aimed at creating fail-safe navigation routes for GPS-independent aircraft.

This research focuses on assessing the feasibility of:

- Identifying geospatially distinctive regions using existing geospatial datasets.
- Quantifying the distribution of such regions across a country or administrative area.
- Developing a systematic methodology to classify these regions based on their recognizability.

Project Aims and Deliverables

Aims

- Develop a methodology to identify visually distinctive regions using OSM data and geospatial analysis tools.
- Analyze the distribution of these regions within a selected administrative area or country.
- Validate the feasibility of using these regions as potential reference points in GPS-denied navigation scenarios.

Deliverables

1. A processed dataset highlighting visually distinctive regions in selected urban or rural areas.
2. Statistical analysis and visualizations of the distribution of these regions within the study area.
3. A final thesis document detailing the methodology, results, and feasibility analysis.

Project Timeline

- **Month 1:** Conduct literature review and extract OSM data for selected areas. Explore metrics such as intersection density, street orientation, and entropy.
- **Month 2:** Develop methods to identify geospatially distinctive regions. Perform initial analysis on selected study areas.
- **Month 3:** Conduct a detailed analysis of region distribution. Visualize the results using GIS tools.
- **Month 4:** Refine the methodology and validate the results. Begin writing the thesis document.
- **Month 5:** Complete the thesis document and prepare defense presentation, including visuals.

References

[1] Geoff Boeing. Osmnx: A python package to work with graph-theoretic openstreetmap street networks. *Journal of Open Source Software*, 2(12):215, 2017.

[2] Geoff Boeing. Urban spatial order: street network orientation, configuration, and entropy. *Applied Network Science*, 4(1), August 2019.

[3] Marc Barthelemy Rémi Louf. A typology of street patterns, 2014. Retrieved from `https://arxiv.org/abs/1410.2094`.